

Note: if I ask you to find

$\iint_D f(x,y) dx dy$, whatever shape D may be
whatever $f(x,y)$ looks like

→ this does not mean you have to go $dx dy$
order is up to you

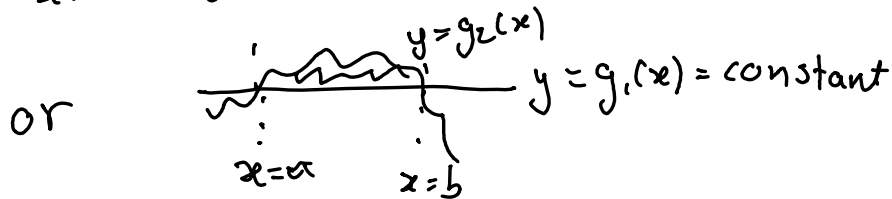
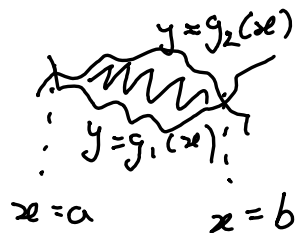
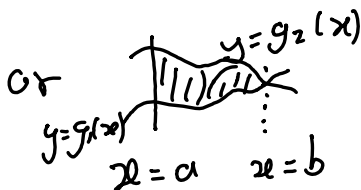
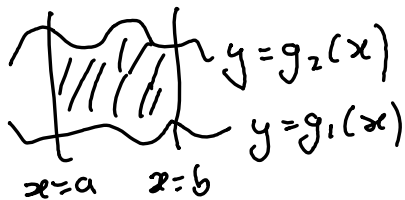
also, side note: $A(D) = \iint_D 1 dx dy$
area

Second Fubini theorem

Let $f: D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ be a continuous function

① if $D = \{(x, y) \in \mathbb{R}^2 \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$ where $g_1, g_2: [a, b] \rightarrow \mathbb{R}$ are continuous f^{ns} s.t. $g_1(x) \leq g_2(x) \forall x \in [a, b]$

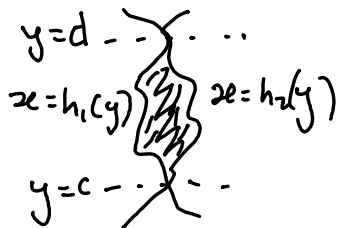
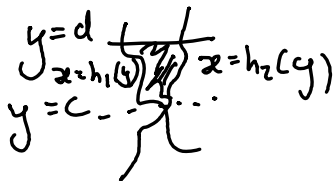
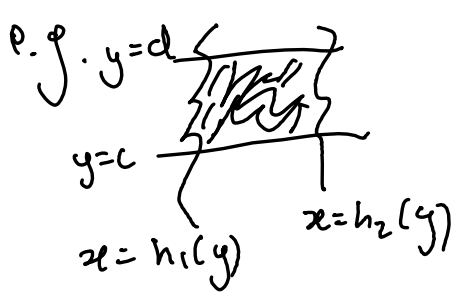
e.g.



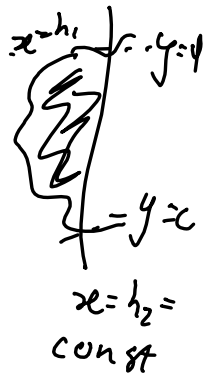
then

$$\iint_D f(x, y) \, dx \, dy = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx$$

② if $D = \{(x, y) \in \mathbb{R}^2 \mid c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$ where $h_1, h_2 : [c, d] \rightarrow \mathbb{R}$ are continuous f^{ns} s.t. $h_1(y) \leq h_2(y) \forall y \in [c, d]$



or



then

$$\iint_D f(x, y) dx dy = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy$$

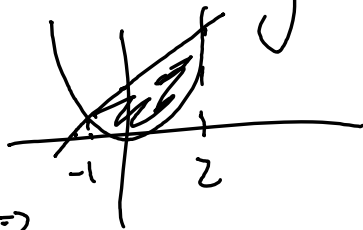
integrate $f(x,y) = x^2 y$. D is above $y = x^2$

below $y = x+2$

intercepts?

$$x+2=x^2 \Rightarrow x^2-x-2=0 \Rightarrow$$

$$(x+1)(x-2)=0 \Rightarrow$$



$$-1 \leq x \leq 2$$

$$\int_{-1}^2 \int_{x^2}^{x+2} x^2 y \, dy \, dx = \int_{-1}^2 x^2 \left[\frac{y^2}{2} \right]_{x^2}^{x+2} dx = \int_{-1}^2 x^2 \left(\frac{(x+2)^2}{2} - \frac{x^4}{2} \right) dx$$

$$= \int_{-1}^2 \frac{x^2}{2} (x^2 + 4x + 4 - x^4) dx = \frac{1}{2} \int_{-1}^2 (-x^6 + x^4 + 4x^3 + 4x^2) dx$$

$$= D \neq y.$$

What about D being between $y = x^2$ and $x = y^2$

if you first do dy :

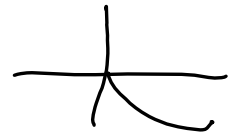
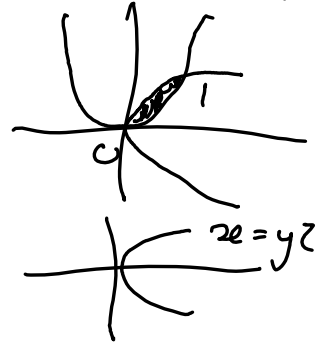
above is $y = +\sqrt{x}$

below is $y = x^2$

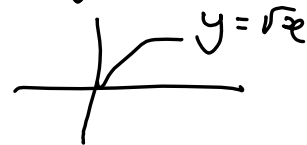
$$0 \leq x \leq 1$$

$$\int_0^1 \int_{x^2}^{\sqrt{x}} f(x, y) dy dx$$

check: is that true @ every pt?
yes!



$$y = -\sqrt{x}$$



First dx ? left is $x = y^2$ right $x = \sqrt{y}$

$$0 \leq y \leq 1$$

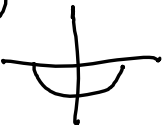
$$\int_0^1 \int_{y^2}^{\sqrt{y}} f(x, y) dx dy$$

What if D is above $y=x^2$ but ^{inside} circle centred at origin, radius $\sqrt{2}$ (i.e. $x^2+y^2=(\sqrt{2})^2=2$)

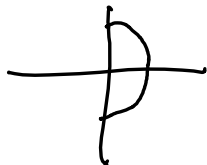
first thing: circle is $x^2+y^2=r^2$



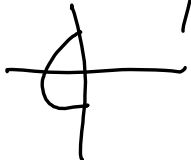
$$y = \sqrt{r^2 - x^2}$$



$$y = -\sqrt{r^2 - x^2}$$



$$x = \sqrt{r^2 - y^2}$$



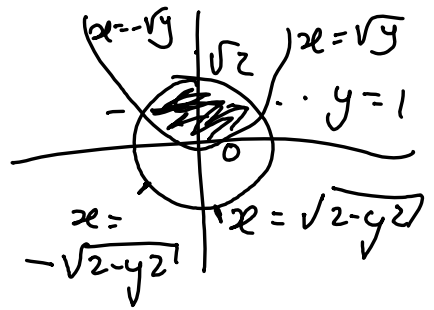
$$x = -\sqrt{r^2 - y^2}$$

so above is $y = \sqrt{2 - x^2}$ below $y = x^2$
intersection? at $(-1, 1)$ and $(1, 1)$. so $-1 \leq x \leq 1$

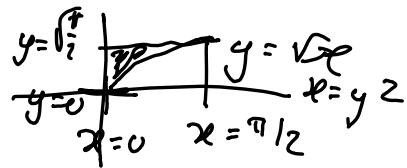
$$\int_{-1}^1 \int_{x^2}^{\sqrt{2-x^2}} f(x, y) dy dx$$

can I do that the other way?
 yes, but will have to split!

$$\int_0^1 \int_{-y}^y f(x,y) dx dy + \int_1^{\sqrt{2}} \int_{-\sqrt{2-y^2}}^{\sqrt{2-y^2}} f(x,y) dx dy$$



consider $\int_0^{\pi/2} \int_{\sqrt{x}}^{\sqrt{\pi/2}} \frac{1}{y} \cos(y^2) dy dx$



can't integrate $\int \frac{1}{y} \cos(y^2) dy$ - at least, not in that order.

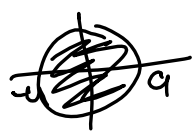
what happens if we integrate, instead $\int_0^{\sqrt{\pi/2}} \int_0^{y^2} \frac{1}{y} \cos(y^2) dx dy$

$$\int_0^{\sqrt{\pi/2}} \frac{1}{y} \cos(y^2) [x]_0^{y^2} dy = \int_0^{\sqrt{\pi/2}} y \cos(y^2) dy \quad (a)$$

notice, $\frac{d}{dy} \frac{1}{2} \sin(y^2) = \frac{1}{2} 2y \cos(y^2)$ so

$$(a) = \left[\frac{1}{2} \sin(y^2) \right]_0^{\sqrt{\pi/2}} = \frac{1}{2} \sin(\pi/2) - 0 = \frac{1}{2}.$$

in circle?



$$\int_0^a \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} f(x,y) dx dy$$

square roots are a pain. other way to describe a circle? $0 \leq r \leq a$ $0 \leq \theta \leq 2\pi$ domain converted
 $f(x,y) = f(r \cos \theta, r \sin \theta)$ since $x = r \cos \theta$ $y = r \sin \theta$
integrand converted.

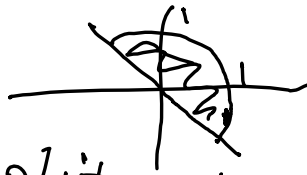
how do I convert $dx dy$? tricky part
 $dx dy = r dr d\theta$

r is the Jacobian for this change of variables

$$\int_0^{2\pi} \int_0^a f(r \cos \theta, r \sin \theta) r dr d\theta$$

notice, no f 's in bounds! nrs!

What about



you'd have to split the domain, and
still deal w/ square roots... or

$$\int_{-\pi/4}^{3\pi/4} \int_0^1 f(r \cos \theta, r \sin \theta) r \, dr \, d\theta$$

we'll talk more about change of
variables next week.